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S2**Started on** Thursday, 24 September 2026, 8:36 AM**State** Finished**Completed on** Thursday, 24 September 2026, 8:47 AM**Time taken** 10 mins 54 secs**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:**Input:** 6**Output:** 6**Explanation:** There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format**Print: The number of possible ways 'n' can be represented using 1 and 3**

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2
3  long long countWays(int n) {
4
5      if (n < 0) return 0;
6      if (n == 0) return 1;
7
8      long long dp[n + 1];
9
10     dp[0] = 1;
11
12     for (int i = 1; i <= n; i++) {
13         long waysWith1 = dp[i - 1];
14         long waysWith3 = (i >= 3) ? dp[i - 3] : 0;
15
16         dp[i] = waysWith1 + waysWith3;
17     }
18
19     return dp[n];
20 }
21
22 int main() {
23     int n;
24
25     if (scanf("%d", &n) != 1) {
26         return 1;
27     }
28
29     printf("%lld\n", countWays(n));
30
31     return 0;
32 }
33

```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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